

Martin Tran

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EXPERIENCE

AWS (AMAZON WEB SERVICES)

Seattle, WA

Software Development Engineer Intern

May 2025-August 2025

- Authored a **detailed technical design document** outlining **CI/CD architecture** and **API integration strategy**; received **positive feedback** from 4+ senior engineers and managers for clarity, scalability, and alignment with team goals
- Designed and implemented a CI/CD pipeline feature to streamline deployment workflows for Amazon developers, improving release reliability and reducing manual effort
- Delivered a **production-ready full-stack service** adopted by Amazon developers, improving developer productivity and accelerating iteration cycles through a React/TypeScript dashboard by displaying the hidden policies
- Partnered with **10+ engineers** across agile sprints, contributing to design reviews, code reviews, and stand-ups to ensure feature alignment with team-wide technical goals
- **Implemented** comprehensive unit and integration tests across all features, achieving **90%+ code coverage** and strengthening reliability in production environments

Tools: Java, Spring Boot, TypeScript, React, Python, Kotlin, AWS DynamoDB, AWS Lambda, JUnit, Mockito

SJSU PEER CONNECTIONS

San Jose, CA

Embedded Tutor - CS 149 (Operating Systems) & Learning Assistant- PHYS 50 (General Physics I - Mechanics)

Jan 2025-Present

- Supporting in-class lecture for CS 149 by circulating during classes, assisting students with questions on challenging operating system concepts
- Providing one-on-one guidance in C and Unix/Linux system calls, improving students' debugging and problem-solving skills
- Tutoring **500+ students** across a wide range of subjects, including CS, Math, Physics, and Computer Engineering, explaining concepts, guiding problem-solving, and preparing students for upcoming exams, and having the **highest average utilization rate** of **89.41%** and the **highest number of total appointments**
- Coordinating closely with the embedded class's professor to enhance student learning, clarify difficult topics, and ensure a strong understanding of CS 149 concepts
- Supported students in a big lecture hall of **200+** people by facilitating weekly small-group, in-class activities that improved engagement and conceptual understanding

PUBLICATIONS & POSTERS

- Tran, M., Indukuri, M., Naseerkhan, E., Rose, J., & Park, Y. (2026, March). *Reinforcement Learning-Based Strategies for Improving CAPTCHAs*. IEEE SVCC 2026.

PROJECTS

REINFORCEMENT LEARNING-BASED CAPTCHA

- Designing a PPO+LSTM RL agent with a 7-action policy for high-security applications, currently achieving 96% accuracy under the supervision of Dr. Younghee Park
- Building an XGBoost classifier achieving 98.5% accuracy by designing an adversarial augmentation pipeline that progressively humanizes bot sessions across 3 difficulty tiers
- Designing a 5-tier adversarial bot framework spanning scripted automation to LLM-powered agents (Claude, Gemini), generating 197 labeled sessions to stress-test detection robustness across increasing behavioral sophistication

Tools: JavaScript, VueJS, React, Python, Flask, scikit-learn, Reinforcement Learning, XGBoost, MySQL

SPAM EMAIL CLASSIFICATION ([Link](#) | [Paper](#))

- Processed and cleaned a dataset of over **10,000** raw emails (SpamAssassin), including regex-based URL/email masking, newline handling, and tokenization
- Developed and trained a transformer-based email spam classifier using **DistilBERT**, achieving **98.8%** accuracy on test data
- Achieved **F1 scores of 0.982 (spam)** and **0.990 (ham)**, demonstrating robust model generalization and balanced class performance

Tools: Python, PyTorch, Hugging Face, scikit-learn, Transformer Model

R.I.C.E PROJECT ([Link](#)) - Financial Budgeting for College Students

- Collaborated effectively with a team of 4 to deliver a high-quality full-stack web application utilizing ReactJS for the front end and Flask for the back end on time and within budget
- Developed and deployed **over 20 robust RESTful APIs** to streamline and enhance user interactions, resulting in a **30% increase** in application performance

Tools: React, JavaScript, Python, Flask, PostgreSQL, Supabase, Tailwind CSS

EDUCATION

San Jose State University

San Jose, CA

Bachelor of Science (B.S.), **Software Engineering** (GPA: 3.6/4.0, Dean's Scholars)

- Activities: SJSU Software and Computer Engineering Association (SCE), Science Extravaganza, MESA

Relevant Coursework:

Java Programming, Object-Oriented Design, Database Management, Data Structures & Algorithms, Software Engineering, Web Development, Computer Organization, Computer Networks, Operating Systems, Machine Learning, Programming Paradigms, Distributed Systems

LEADERSHIP/INVOLVEMENT

SJSU SOCIETY OF COMPUTER & SOFTWARE ENGINEERING (SCE)

San Jose, CA

EP/PR Officer - Industry Lead

January 2024-December 2025

- Coordinated and managed outreach to 10+ tech companies and successfully executed logistics for **3+ industry site tours**, connecting SJSU students with real-world engineering environments and professionals.

SKILLS

Programming: Java, Python, JavaScript/TypeScript, SQL/NoSQL, C/C++

Technologies: Linux, Git/GitHub, React.js, Flask, SpringBoot, Tailwind CSS, AWS Services, MySQL, PostgreSQL, SQLite, GraphQL, PyTorch/TensorFlow